DATAMITES

Rubiex Internship Project — Data Science Capstone Project

PRCP-1009: Cellphone Price Range Prediction Using Machine Learning

Executive Summary Report

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Team Code / Project Code	PTID-AIE-JUL-26-11142 / PRCP-1009
Dataset	Mobile Price Classification — 2,000 records, 20 features
Best Model	Logistic Regression (Tuned) — 98.25% Accuracy

1. Problem Statement & Objective

The objective of this project is to analyze mobile phone specifications and build a machine learning classification model to predict the price range of a mobile phone — Low, Medium, High, or Very High — based on its technical features. The project compares multiple classification algorithms, identifies the most influential price-driving features, and delivers business insights for effective pricing decisions.

2. Dataset & Methodology

The dataset (datasets_11167_15520_train.csv) contains 2,000 mobile phone records with 20 input features — including battery power, RAM, internal memory, camera resolution, processor cores, screen dimensions, and connectivity options — and a balanced target variable (price_range) with 500 records in each of the four classes. The data was found to be clean, with no missing values or duplicate records.

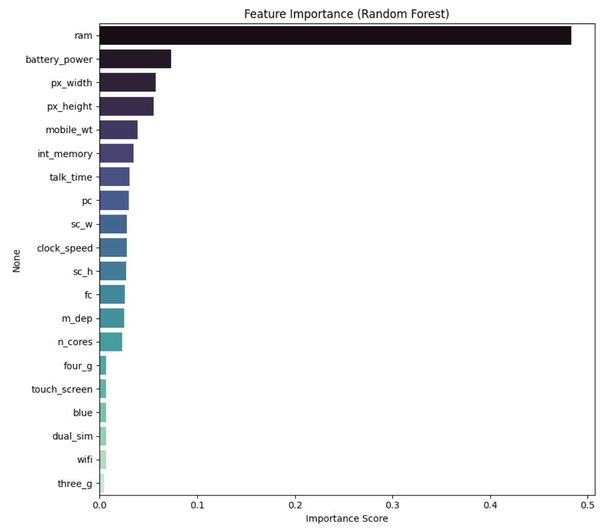
The dataset was split 80:20 into training and test sets with stratified sampling. Features were scaled using StandardScaler for distance-sensitive algorithms (Logistic Regression, KNN, SVM), while tree-based models used raw features. Six classification algorithms — Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting — were trained and evaluated, and the best model was further optimised using GridSearchCV hyperparameter tuning.

3. Model Comparison & Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression (Tuned)	0.9825	0.9826	0.9825	0.9825
Logistic Regression	0.9650	0.9650	0.9650	0.9650
Gradient Boosting	0.9125	0.9124	0.9125	0.9123
SVM	0.8950	0.8969	0.8950	0.8956
Random Forest	0.8775	0.8776	0.8775	0.8774
Decision Tree	0.8300	0.8319	0.8300	0.8302
KNN	0.5175	0.5267	0.5175	0.5199

After GridSearchCV tuning (best parameters: C=100, solver='lbfgs'), Logistic Regression improved from 96.50% to 98.25% accuracy, making it the recommended production model. It offers the highest accuracy, fewest misclassifications, and strong computational efficiency compared to the ensemble and distance-based alternatives tested.

4. Key Findings — Feature Importance



Feature Importance (Random Forest)

- **RAM:** By far the strongest predictor (~48% importance / 0.917 correlation with price range).
- **Battery Power:** Second most influential specification (~7% importance).
- **Screen Resolution (px_width, px_height):** Moderately influential; higher resolution associates with higher price tiers.
- **Internal Memory:** Meaningful but secondary influence on price range.
- **Connectivity (Bluetooth, Wi-Fi, Dual SIM, 3G):** Minimal influence — largely standardised across all price segments.

5. Business Insights

- Data-Driven Pricing: The model enables manufacturers to estimate the appropriate price range from a device's specifications rather than relying on manual estimation.
- Product Planning: Manufacturers can optimise specifications for target price segments without unnecessary cost increases.
- Competitive Positioning: The model supports benchmarking planned specifications against market price trends.
- Reduced Pricing Errors: Machine-learning-based estimation minimises the risk of overpricing or underpricing new devices.

6. Conclusion

This capstone project successfully developed and validated a machine learning pipeline capable of classifying mobile phones into four price categories with 98.25% accuracy using a tuned Logistic Regression model. RAM, battery power, and display resolution were identified as the primary drivers of mobile phone pricing. The findings provide manufacturers with a reliable, data-driven decision-support tool for product specification planning and competitive pricing strategy. Future enhancements could include larger and more diverse datasets, additional real-world features such as brand and market trends, and deployment of the model as a live web application or API.

7. Project Links

GitHub Repository: github.com/Sameekshagithub/PRCP-1009-CellphonePrice-Prediction-Using-Machine-Learning-Classification-